YMNOTES MEDICAL

Hernias and Inguinoscrotal Swellings

REDESIGNED EDITION

About this edition

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Presentation only. The medical text is reproduced from the original book: typography, hierarchy, page layout, tables, figures and navigation were rebuilt; the wording, numbers, doses, units, thresholds, classifications and recommendations were not.

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Typographical defects in the source — misspelled drug names, malformed medical terms, lost word spaces — were corrected where the evidence was unambiguous, and marked where the correction changes a word. Nothing was rewritten for style, and no medical statement was modernised: where the source is dated rather than wrong, it stands as written.

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How to read this book

CHAPTER OPENER

Each chapter opens on its own page with the chapter number and the original book's own title for that material.

TOPIC BAND

A tinted band with a heavy left rule marks a major topic. The running header repeats the current topic on every page.

CONDITION

A named disease, drug or entity sits under a lighter rule directly beneath its topic.

RECURRING SECTIONS

Definition, Causes, Clinical Picture, Investigations, Treatment and Complications carry a small accent marker. Treatment runs wide and dark; complications and contraindications run red; investigations run blue.

BULLET DEPTH

A filled diamond is a first-level point, a filled square a second, a short dash a third and a hollow circle a fourth. Shape carries the depth — not size or colour — so the nesting survives greyscale printing and reads at arm's length.

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SOURCE PAGES

Figure captions cite the page of the original book they came from, so any passage can be traced back to the source.

CHAPTER 01

Abdominal wall Hernias

IN THIS CHAPTER

01 Definition

02 Types of hernias

03 External abdominal wall hernias

01

Definition:

It is protrusion of a viscus or part of a viscus within a peritoneal sac through a defect. So for a hernia to occur there must be a **defect – sac –viscus**.

Types of defects in hernias:

1- Natural→ as internal ring – femoral ring- oesophageal hiatus

2- Acquired → as weakness of posterior inguinal wall in direct hernia.

Types of sacs:

1- Congenital sac = preformed = dated since birth e.g. a) congenital inguinal hernia due to patent processus vaginalis

b) Indirect inguinal hernia in infants or young adults.

2- Acquired sac:

developed due to **raised intra-abdominal pressure** (pulsion sac) as

a) Indirect inguinal hernias in adults or old ages

b) Direct inguinal hernias

c) Femoral hernias

d) PUH – Incisional hernias

Types of hernias:

1- External abdominal wall hernias

2- Internal hernias as hiatus hernia

External abdominal wall hernias

Types:

A- Midline hernias:

Epigastric hernia - fatty hernia of linea alba –PUH- Umbilical hernia – Incision hernia.

B- Lateral hernias:

Inguinal hernias – femoral hernia –incisional hernias.

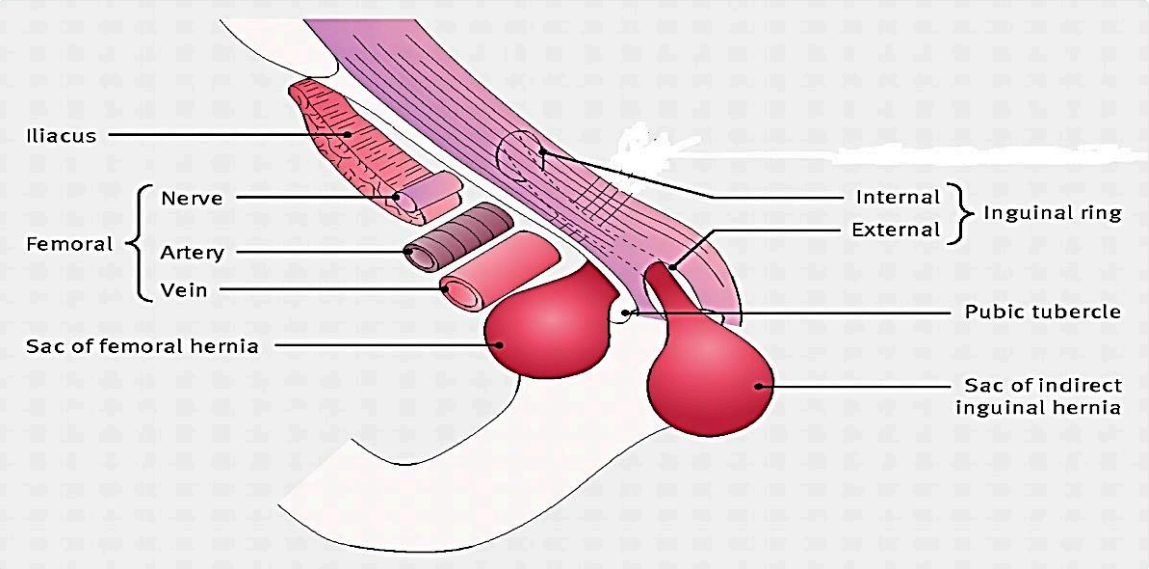


Fig 1.1 — original page 2

CHAPTER 02

Inguinal hernias

IN THIS CHAPTER

01 Types
02 Factors that prevent development of inguinal hernias
03 Indirect Hernia
04 Direct Hernia
05 Clinical picture of inguinal hernias
06 Different types of inguinal hernia
07 Investigations of hernia
08 Treatment of inguinal hernia
09 General treatment
10 Surgical treatment
11 Indirect hernia
12 Hernioplasty or herniorrhaphy
13 Direct hernia

02

It is type of lateral abdominal wall hernias that occurs in the groin areas

Site: inguinal canal above the medial half of the inguinal ligament.

Types:

1-Indirect inguinal hernia

2-Direct inguinal hernia

Factors that prevent development of inguinal hernias:

a) Strong posterior wall (fascia transversalis & conjoint tendon)

b) The internal and external rings present in 2 levels (ant. & post. Walls)

c) Shutter mechanism of the inguinal canal (i.e. during straining → approximation of conjoint & inguinal ligament → augments the posterior wall.

d) Internal oblique support internal ring and conjoint tendon support the external ring.

Indirect Hernia

I- Aetiology &

Pathogenesis:

1- The sac may be:

a- Congenital preformed sac as patent part or whole processus vaginalis. **b- Acquired sac** : on top of raised intra-abdominal pressure

(chronic bronchitis- chronic constipation- ascitis- lifting heavy objects)

2- predisposing factors:

chronic strain →increased intra- abdominal pressure as in ..... a-↑ intra-abdominal pressure →acquired sac →protrude through internal ring inside the inguinal canal + viscus or omentum = indirect hernia.

b- Congenital preformed sac in infant + strain = congenital inguinal hernia. C- Congenital sac in adult + predisposing factor of long duration = indirect inguinal hernia.

3-Defect:

internal ring (1/2 inch above mid-inguinal point).

- **II- Content of the hernia:**

1- Omentum (doughy, no gurgling, easily introduced early & difficult late)

2- Any organ or viscus except pancreas (as it is fixed retroperitoneal organ).

Intestine (soft, + gurgling, 1st part difficult to be reduced then easy late).

III- Coverings of the hernia:

from superficial to deep:

- Skin, superficial fascia, E.O. aponeurosis, external spermatic fascia, cremasteric muscle & fascia and internal spermatic fascia.
- External spermatic fascia taken at the external ring.
- Cremasteric muscle & fascia taken from internal oblique muscle inside the canal.
- Internal spermatic fascia taken from fascia transversalis at internal ring.

Direct Hernia

I- Aetiology &

Pathogenesis:

1- The sac:

is only Acquired sac on top of raised intra-abdominal pressure

(chronic bronchitis- chronic constipation - ascitis- lifting heavy objects- enlarged prostate).

2- predisposing factors:

chronic strain →increased intra-abdominal pressure as in .....

- ↑Intra-abdominal pressure → weakness of the fascia transversalis (post. Wall) → pushed by peritoneal sac inside the inguinal canal postero- anterior + viscus or omentum inside = **Direct Hernia.**

3- Defect:

fascia transversalis **(Hesselbach's triangle)** or conjoint tendon.

Hesselbach's triangle:

bounded by

1- Rectus abdominis →medial

2- Inferior epigastric vessels →lateral

3- Inguinal ligament → inferiorly.

4- Floor → fascia transversalis.

- Importance: it can differentiate intra-operatively direct hernia sac from indirect sac → the direct sac is medial while the indirect is lateral to the vessel.

II- Content of the hernia → same as indirect hernia

III- Coverings of the hernia:

the sac lies posterior to the cord

From superficial to deep → the same as indirect hernia except that not covered by external spermatic fascia.

Clinical picture of inguinal hernias:

- **General symptoms:**

- **Age**: infant → congenital hernia
- Young adult → usually congenital hernia
- Adult & old → direct or indirect hernia.
- **Occupation**: lifting heavy objects
- **Associated disease**: Chronic cough – Chronic constipation- strain at micturition (old prostatic patient). Liver disease (ascitis).

Local symptoms:

- **Rt. or Lt. groin swelling:**
- Reducible in non-complicated cases
- Long standing cases may reach to the scrotum (indirect).
- Urinary symptoms in sliding hernia (U.bladder).
- Symptoms of obstruction in obstructed hernias.
- Symptoms of obstruction & peritonitis in strangulated cases.

General signs:

chest for bronchitis

Abdominal:

Ascitis – other hernia orifices

P/R:

in suspected enlarged prostate.

- **Local examination:**

1- Swelling in the inguinal region gives expansile impulse on cough.

2- The swelling is above the medial half of the inguinal ligament.

3- The swelling is reduced antero-posteriorly in **direct hernia** and reduced upwards, backwards and laterally in **indirect hernia.**

4- Internal ring test positive in indirect hernia.

- **5- Scrotal neck test:**

a)- swelling above → pure inguinal (direct or indirect H.) b)- swelling is felt in between → inguino-scrotal hernia (indirect H.) c)- swelling below → pure scrotal (hydrocele).

Different types of inguinal hernia

The hernia may be:

- **Complete:** When the hernial sac reaches up to the bottom of scrotum and the testis cannot be felt separately.
- **Incomplete hernia**: When the hernial sac does not descend up to the bottom of the scrotum. This can be:
  - **Bubonocoele**: When the hernial sac is confined to the inguinal canal and does not reach beyond the superficial inguinal ring.
  - **Funicular:** When hernial sac goes beyond the superficial inguinal ring and reaches up to the upper pole of the testis.

Investigations of hernia:

- Chest X- Ray → bronchitis
- Abdominal U/S → in ascitis or abdominal tumours.
- Pre-operative lab. Investigations (CBC., liver & renal functions).

Treatment of inguinal hernia:

I- general treatment:

a)- Control of all predisposing factors before surgery b)-in bilateral hernias we start with the symptomatic or complicated one.

II-Surgical treatment:

A- Indirect hernia:

1- Dissection of the hernia sac within the cord till its neck at the internal ring (known by presence of pre-peritoneal fat).

2- Opening of the sac at the fundus → reduction of the contents inside the abdomen 3- Transfixion and excision of the sac at the neck. **(herniotomy)**

4- Narrowing of the internal ring to admit one finger.

5- Hernioplasty or herniorrhaphy:

I- Hernioplasty:

- Include application of prolene mesh to the posterior inguinal wall → fixed to the conjoint and inguinal ligament by stitches.
- The fibrotic reaction will augment the posterior wall. (The best).

II-Herniorrhaphy:

was used before:

a)-

Bassini repair:

the inguinal ligament is sutured to the conjoint tendon. It is associated with tension over the tissues of the canal.

b)-

Mc-Vay repair:

the conjoint tendon is sutured to the ilio-pubic tract which is a thickening of the fascia transversalis behind the inguinal ligament c)- **Shoudice**

repair:

double breasting of the fascia transversalis stretching and strengthening it.

B- Direct hernia:

1- No herniotomy

2- Hernioplasty →the same as above (the best).

3- Herniorrhaphy→ the same as above.

CHAPTER 03

Femoral hernia

IN THIS CHAPTER

01 Clinical picture of femoral hernia

02 What is low repair for femoral hernia?

03 Low operation technique

03

What is the incidence of femoral hernia?

20 % of hernia in women and 5 % of hernia in males are femoral hernia.

Through which anatomical defect femoral hernia emerges?

The hernia emerges through the Hesselbach's triangle and then finds its way out through the femoral ring into the femoral canal and descends into the thigh through the saphenous opening.

What is the boundary of femoral canal ?

- Femoral canal is a tunnel about 1.25 cm long and 1.25 cm wide at its base
- This is the most medial compartment of the femoral sheath
- This extends from femoral ring above to the saphenous opening below
- Laterally separated from the femoral vein by a septum
- The femoral canal is closed above by septum crurale and below by cribriform fascia.

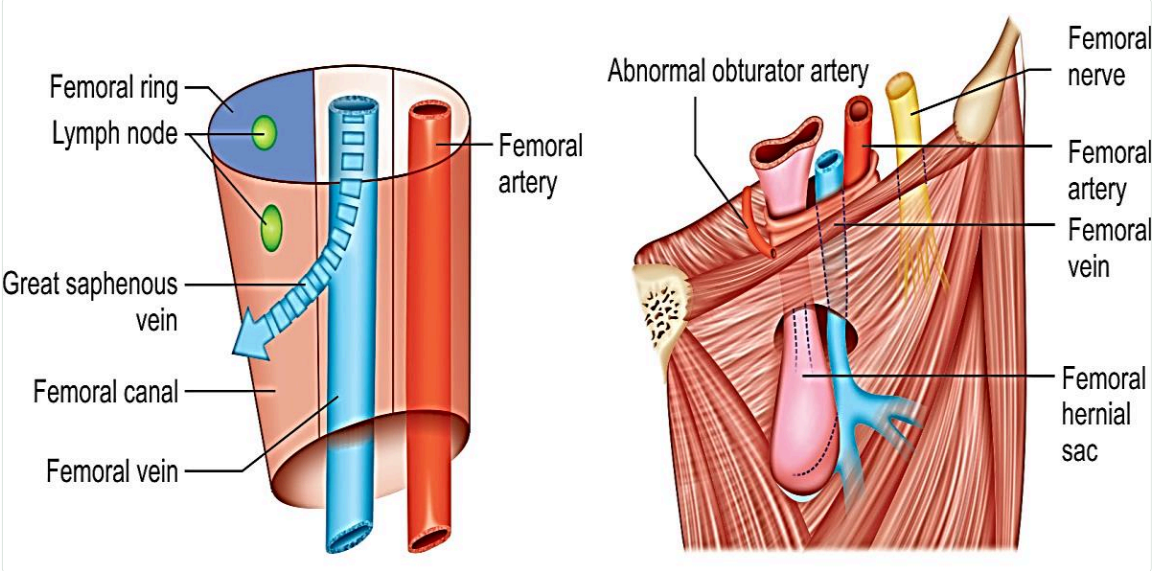


Fig 3.1 — original page 8

Boundaries of femoral sheath

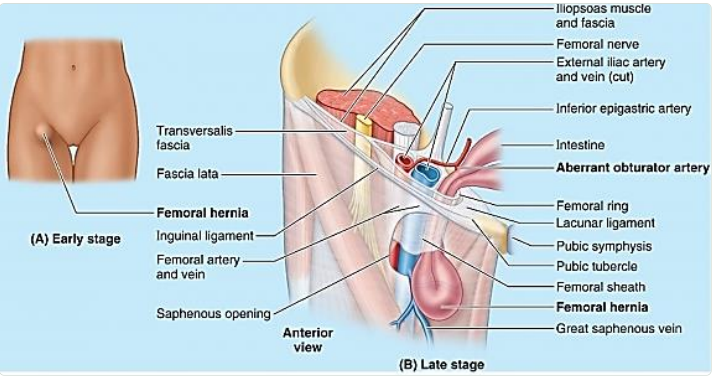


Fig 3.2 — original page 9

- **What are the contents of femoral canal?**
  - It contains a deep inguinal lymph node (Cloquet's lymph node) and lymphatics and loose areolar tissue.
- **What is the boundary of femoral ring?**
  - **Anteriorly:** Inguinal ligament
  - **Posteriorly:** Cooper's ligament & pubic bone
  - **Medially:** Lacunar ligament
  - **Laterally:** By a septum separating the femoral vein.
- **What are coverings of femoral hernia?**



Fig 3.3 — original page 10

- **What are the sites of constriction in strangulated femoral hernia?**

The site of constriction may be at:

1. Lacunar ligament
2. Neck of the sac.

Clinical picture of femoral hernia:

1. Lump in groin or femoral region. With typical signs of a hernia (swelling reducible backwards, soft with impulse on cough)

2- Tender irreducible swelling in the groin.

2. Acute intestinal obstruction (pain, vomiting, distension + swelling in the Upper thigh)

3. Strangulated hernia with pain, vomiting, tender, irreducible swelling in groin Or thigh.

- **What are the differential diagnosis of a femoral hernia?**

1. Inguinal hernia (how to differentiate ?)
2. Saphena varix
3. An enlarged Cloquet's node
4. Lipoma
5. Femoral aneurysm
6. Psoas abscess pointing into the thigh.

- **Why femoral hernia needs early operation?**

The incidence of strangulation in femoral hernia is very high.

About 30 – 45% of femoral hernia may undergo strangulation in 2 years' time.

The narrow and its tight femoral ring predispose to strangulation.

- **What operation will you do?**

consider mesh repair of femoral hernia for this patient.

What is low repair for femoral hernia?

Low operation technique

- The incision is located in the medial aspect of the thigh below the inguinal ligament centering over the femoral canal
- The subcutaneous fat is incised to expose the hernial sac
- The hernial sac is dissected all around the inguinal ligament, and Cooper's ligament
- The sac is opened, contents inspected and reduced into the peritoneal cavity.
- Neck of the sac : If the neck of the sac is very tight—lacunar ligament may need to be incised to facilitate reduction of hernia content.
- Abnormal obturator artery is to be taken care of while incising the lacunar ligament
- The sac is ligated by transfixation at the neck and redundant sac excised
- The repair is done with 2-0 polypropylene suture apposing the inguinal ligament to Cooper's ligament with few interrupted suture or by a purse string suture closing the femoral ring.

CHAPTER 04

Midline hernia

IN THIS CHAPTER

01 Epigastric hernia & Fatty hernia of linea alba

02 Epigastric hernia

03 Fatty Hernia Of Linea Alba

04

Epigastric hernia & Fatty hernia of linea alba

I- Epigastric hernia:

= true hernia

Site:

midline epigastric region

Aetiology :

weakness of the linea alba due to obesity, raised intra-abdominal pressure,....

Coverings:

skin and pre-peritoneal fat

Sac:

acquired peritoneal sac

Defect:

linea alba

Contents:

intestine or omentum.

II-Fatty Hernia Of Linea Alba

= Not true hernia

It is protrusion of the pre-peritoneal fat through a defect in linea alba without sac

Clinical picture:

Swelling in the epigastric region usually irreducible

Sometimes dyspepsia is present

Local examination: epigastric swelling with expansile impulse on cough (in epigastric H.)

Investigation:

the same

Treatment:

1- Mayo's repair or 2- Hernioplasty

CHAPTER 05

Umbilical hernias

IN THIS CHAPTER

01 Types

02 Adult para-umbilical hernia

05

Types:

- 1- Congenital umbilical hernia = Exomphalos
- 2- Infantile umbilical hernia
- 3- Adult para- umbilical hernia

Adult para-umbilical hernia

- **Aetiology:** more common in females – obese - repeated pregnancy
- **Defect:** linea alba above the umbilicus
- **Sac:** acquired multi-locular
- **Coverings:** skin, fascia & pre-peritoneal fat.
- **Contents:** usually omentum
- **Clinical picture:** same as before –the swelling is always above the umbilicus due thinner & wider linea alba.



Fig 5.1 — original page 13

- **Investigations:** As before + abdominal ultrasound

Treatment:

I- Hernioplasty:

- Dissection of the sac → open at the neck →reduction of the content →excision of the sac → closure of the defect →prolene mesh

II- Mayo's repair:

The same as above, but the defect is closed by overlap the upper flap over the lower flap of rectus sheath.

CHAPTER 06

Incisional and recurrent hernias

IN THIS CHAPTER

01 Causes
02 Patient related factors
03 Pre-operative factors
04 Operative factors
05 *In Incisional hernia
06 Post-operative factors
07 Recurrent hernia
08 Incisional hernia
09 Hernioplasty
10 Anatomical repair

06

Causes:

A- Patient related factors:

1- No treatment of the predisposing factors (chronic strain).

2- Weak abdominal wall muscles

3- Marked obesity

4- Diseases that lower patient's immunity →predisposes to infection as D.M., liver cell failure, steroid therapy,...

B- Pre-operative factors:

the same as above.

C- Operative factors:

*

In Recurrent hernia:

1- Incomplete removal of the sac

2- Missed sac (in pantaloon hernia).

3- Bad handling of the tissues with much trauma→increases incidence of infection & ischemia of the tissues.

4- Bad choice of repair (herniorrhaphy associated with more recurrence than hernioplasty).

5- Use of absorbable sutures in herniorrhaphy.

*In Incisional hernia:

1- Bad handling of the tissues with much trauma→increases incidence of infection & ischemia of the tissues.

2- Muscle cutting incision is more liable for incisional hernia than muscle splitting.

3- Use of absorbable sutures in closure of the abdominal wall.

D- Post-operative factors:

1- Persistent increase in the intra-abdominal pressure post-operatively.

2- Wound hematoma and infection → weakness of the tissues.

Clinical picture:

- Typical signs and symptoms of hernia present in relation to scar of previous hernia repair (recurrent H.) or abdominal wound.
- The hernia may be reducible or partially irreducible due to presence of adhesions.
- Complications of hernia are common due to adhesions.
- The defect of the abdominal wall may be felt especially in incisional hernias.



Fig 6.1 — original page 15

Investigations:

same as before.

Treatment:

A- Recurrent hernia:

1- Correction of the predisposing factors.

2- **Hernioplasty** is the best.

B- Incisional hernia:

1- Hernioplasty:

Dissection of the hernia sac →reduction of the sac inside the abdomen → closure of the defect →by non-absorbable sutures (prolene).→ Then on lay prolene mesh applied to the abdominal wall.

2- Anatomical repair:

we repair the defect in layers according to the site of incision.

CHAPTER 07

Sliding hernia

IN THIS CHAPTER

01 Clinical picture

02 Treatment

07

- It is hernia where a viscus form part of the wall of the sac not a content.
- Commonest sliding organ is urinary bladder, cecum and sigmoid colon.

Clinical picture:

like before +

1- The hernia usually indirect, longstanding, partially irreducible and may reach the scrotum

2- Urinary symptoms may be present as double micturition.

Treatment:

hernia management as before

The sliding organ is pushed back to the abdomen.

CHAPTER 08

Congenital hernias

IN THIS CHAPTER

01 Congenital inguinal hernia
02 Congenital umbilical hernias
03 Exomphalos minor
04 Exomphalos major
05 Congenital umbilical hernia
06 Clinical presentations
07 What operation will you do?

08

I- Congenital inguinal hernia

2- Congenital umbilical hernia

Congenital inguinal hernia

- Site of the swelling—in one or both groins (inguinal) or inguino-scrotal dated since birth or shortly after.
- the swelling increases when the child strains (coughing, crying) and reduced or disappeared on lying down
- sometimes there is pain over the swelling
- A history of irreducibility of the swelling may be present
- history suggestive of acute intestinal obstruction associated with irreducibility of the swelling (abdominal pain, vomiting, absolute constipation and abdominal distension).
- Any history of chronic straining like—chronic cough, constipation or difficulty in passing urine.

Examination—Inspection/Palpation

- Congenital hernia usually extends up to the bottom of the scrotum (inguino-scrotal).
- Expansile impulse on cough is evident.
- **Scrotal neck test:** Whether it is possible to get above the swelling (swelling is scrotal or inguino-scrotal ?) (Pure scrotal = congenital hydrocele)
- Reducibility—an uncomplicated hernia is reducible
- **Deep ring test**— congenital hernia occurs through the deep ring. a deep ring test is not required in examination of congenital hernia as inguinal canal is not developed in infants.
- Assess the content of the hernia sac—omentum or intestine. Doughy feel, dull note on percussion and absent bowel sounds will suggest omentocele

  Soft elastic feel, reduction with gurgling sound, resonant percussion note and presence of bowel sound will suggest enterocele.

How is the inguinal canal in children?

In newborn, the inguinal canal does not develop and superficial and deep inguinal rings lie close to each other. By 2 years the deep ring moves laterally and proper inguinal canal can be identified.

What is the basic defect in congenital hernia?

The basic defect in congenital hernia is the failure of obliteration of processus vaginalis. The inguinal canal structure and mechanism remain intact.

What is the chance of developing hernia on opposite side?

About 10% of children on follow-up develop contralateral hernia.

Why early operation is indicated in infants with congenital hernias?

Early operation is indicated in infants because of high chance of strangulation.

What operation will you do?

The surgical treatment will be **herniotomy.**

What is the difference in technique in older children?

- In new born and young infants direct access to the hernia sac without opening of the canal.
- In child more than 3 years of age, the external and internal rings become widely separated so that direct dissection is no longer possible.
- In these cases, skin incision is made slightly laterally over the internal inguinal ring.
- The skin and superficial fascia is incised. The external oblique aponeurosis is incised and inguinal canal exposed—the external ring is not opened.
- The cremesteric fascia and internal spermatic fascia is incised and sac dissected free from the cord structures.
- The sac is clamped across and the distal part of the sac transected and kept open.
- The proximal part of the sac is dissected up to the deep ring and transfixed.

Congenital umbilical hernias

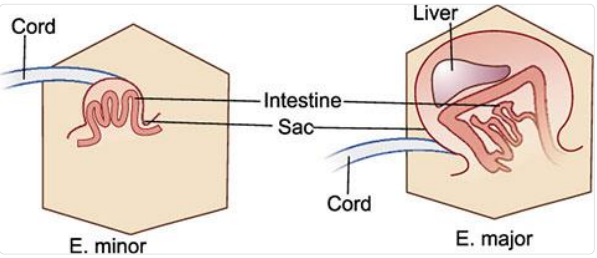


Fig 8.1 — original page 18

I- Exomphalos minor

- When the umbilical defect is less than 4 cm.
- This is also termed as herniation of the umbilical cord and a loop of small gut may remain in the sac and may be injured if umbilical cord is ligated and divided
- In the usual way. The umbilical cord is attached to the top of the sac
- The sac covering the exomphalos may rupture
- During birth, the un-ruptured sac is transparent and is lined by outer amniotic membrane and the inner peritoneum.
- Treatment of exomphalos minor: the small defect may be closed spontaneously soon after birth or surgically.

II-Exomphalos major:

- When the fascial defect in the abdominal wall is more than 4 cm, this is called exomphalos major.
- The sac contains a large part of the small intestine and a part of the large intestine.
- The liver, spleen and pancreas may be seen through the transparent membrane
- **Operative techniques for management of exomphalos major**

1.Primary Closure:

The skin is dissected away from the fascial margin. The abdominal wall is stretched in all the quadrants manually. The herniated viscera are then replaced into the abdominal cavity and the fascial defect apposed in the midline.

2.Staged repair:

The skin margin is freed from the fascial attachments and a mesh is sutured circumferentially in the margin of the fascial gap to cover the viscera. Later on the fascial margins can be apposed with interrupted sutures and the skin closed over the gap.

Congenital umbilical hernia

This is a herniation through a weak umbilical scar which fails to close after birth.

Predisposing factors:

1- Weak umbilical stump (pre-mature babies)

2- Infection of the umbilical stump

3- Persistent increased abdominal pressure due to straining

(cough, constipation, distension)

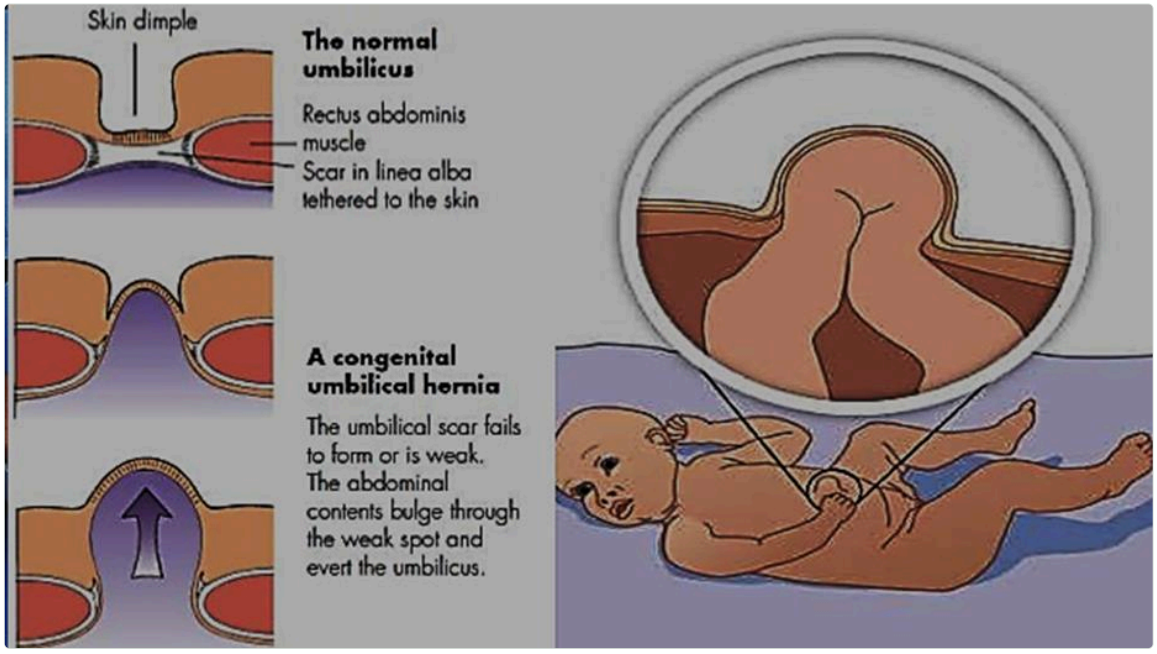


Fig 8.2 — original page 20

Clinical presentations:

- A child presented with a swelling in the umbilicus since birth. The swelling is increasing in size .
- The swelling aggravated by activities like, walking, running and crying
- The swelling spontaneusly reduced on lying down.
- Usually No pain over the swelling
- History of irreducibility may be present.



Fig 8.3 — original page 20

- **On examination:**
- The umbilicus is stretched and everted due to a swelling in the umbilicus.
- The swelling shows expansile impulse on cough
- The swelling reduced spontaneously on lying down.
- There is sometimes a gurgling sound during reduction of the swelling.
- A gap of more or less than 2 cm is palpable in the umbilical scar.

What operation will you do?

a) A curved incision is made below the umbilicus with convexity downwards and the anterior rectus sheath is exposed

b) The skin flap is raised above and the hernial sac is dissected free.

c) The hernial sac is opened and the contents reduced

d) The redundant sac is excised and the hernial sac is closed at the neck

e) The gap in the linea alba is approximated by interrupted absorbable suture

CHAPTER 09

Hernia complications

IN THIS CHAPTER

01 Irreducibility
02 Intestinal Obstruction
03 Strangulation
04 Causes of strangulation
05 Constricting agent
06 Pathology of strangulated hernia
07 Clinical picture of strangulated hernia
08 General
09 Urgent exploration of the hernia

09

I-Irreducibility II- Obstruction

III- Strangulation IV- Inflammation

V- Hydrocele

I-Irreducibility:

Def.: Failure of the contents to be reduced back to the abdomen.

Aetiology:

1- Sudden strain → over-crowding of the contents inside the sac → failure of reduction.

2- Adhesions between the contents and the sac.

Clinical picture:

Same as before +

- History of reducible hernia that becomes irreducible.
- Pain or discomfort related to the hernia.
- No obstruction
- The hernia is not tense.

Investigations:

of hernia as before

Treatment: as any hernia.

II- Intestinal Obstruction

It is occlusion of the intestinal lumen within the hernia.

Causes:

1 - Adhesions between the sac and contents

2 - Sharp edge of the defect constricts the intestine from outside (internal ring in indirect hernia & lacunar ligament in femoral hernia).

3 – Overcrowding of the contents within the sac

The blood supply of the contents are intact.

Clinical picture:

Local:

same as any hernia + Irreducible, tender hernia & not tense.

Generally:

signs & symptoms of intestinal obstruction (**mention**).

Investigations:

- For hernia as before
- Plain X- ray abdomen erect & supine → multiple air and fluid levels and dilated loops.

Treatment:

- Urgent surgery to avoid strangulation of the contents.
- Correction of general condition (Ryle tube, I.V. fluids,...)

(Mention in details)

- Hernia surgery as before ----> (**mention**)
- Before reduction of the contents it should be examined for viability (normal colour, luster, and tone) → allowed for reduction inside the abdomen.
- If Compromised loops , increase warmth, increase oxygen flow if the colour return normal reduce inside the abdomen ; if not resection is done.

III- Strangulation

- Means impairment of the blood supply of the hernia contents
- Gangrene occurs within 6 hours.

Incidence:

Femoral hernia → 30 % of femoral hernias

Inguinal hernia → 4% of inguinal hernias

PUH → 20 % of PUHs

Commonest strangulated hernia is inguinal hernias 50%

(Commonest hernia).

Causes of strangulation:

1-Neglected obstructed hernia

2- Overcrowding of the contents with sharp narrow defect

3- Repeated attempts of reduction of irreducible hernia producing edema of the tissues → more compression.

Constricting agent:

1-The edge of the defect according to the type of the hernia.

2- Constricting adhesions within the sac occluding the intestine from outside.

Pathology of strangulated hernia:

a- Irreducibility of the hernia contents b- Compression of the venous drainage of the content at the site of constriction

→ congestion and edema of the contents→ accumulation of gas and fluids inside the lumen and sac of the hernia.

c- Compression of the arterial supply of the contents →devitalization of the contents → extravasation of toxic fluid & bacteria inside the sac. d- Gangrene of the ischemic contents → perforation of the bowel → peritonitis → septic shock →death.

Clinical picture of strangulated hernia:

General:

1- Picture of strangulation:

dehydration, hypovolemic shock

(dehydration, tachycardia, hypotension &disturbed level of conscious).

2- Picture of peritonitis:

FAHM

- **Abdominal: signs of obstruction + signs of peritonitis (Mention)**

(No obstruction in presence of omentum, Richter's H. & Littre's H.)

Local:

the hernia shows irreducibility – oedema of the skin overlying - no impulse on cough- tense- tender.

Investigations:

1- To diagnose intestinal obstruction →mention??

2- To diagnose strangulation & peritonitis → CBC & Abdominal U/S (for peritoneal collection).

Treatment:

1- Immediate resuscitation: Ryle's tube, I.V. fluids, blood transfusion & I.V. antibiotics

2- Urgent exploration of the hernia:

a- open the fundus of the sac first → suction of the toxic fluids out before division of the constricting ring to avoid escape of the fluids inside the peritoneum. b- Divide the constricting ring and examine the contents:

I-Viable → reduced inside the abdomen followed by hernia repair

II-Non-viable (loss of luster, loss of tone, cyanosed, bluish or blackish & no pulsation of the blood supply) →

resection:

- Omentum → Excised
- Small intestine → resection & anastomosis
- Right colon → Right hemicolectomy + ileo-transverse anastomosis
- Left colon → resection + proximal colostomy → re-anastomosis after 4- 6 weeks.

III- Doubtful loops → hot foments over the loops + increases O2 flow →if turned viable reduce it inside the abdomen while if remained doubtful resect it as above

CHAPTER 10

Hydrocele

IN THIS CHAPTER

01 Types
02 Primary hydrocele
03 Secondary hydrocele
04 Inguinal herniorrhaphy?
05 Q: What is hydrocele of the hernial sac?
06 Q: What is chylocele?
07 Q: What is hematocele?
08 Types of hydroceles
09 Congenital hydrocele
10 Funicular hydrocele
11 Infantile hydrocele
12 Encysted hydrocele of the cord
13 Vaginal hydrocele
14 Bilocular hydrocele or hydrocele en bisac
15 Hydrocele of the hernial sac
16 Q: What are the characteristics of fluid in hydrocele?
17 Clinical picture of vaginal hydrocele
18 D.D. of hydrocele
19 Scrotal ultra-sound
20 Eversion of tunica vaginalis
21 Lord's operation
22 Encysted hydrocele of the cord

10

Hydrocele is a collection of fluid in the tunica vaginalis .

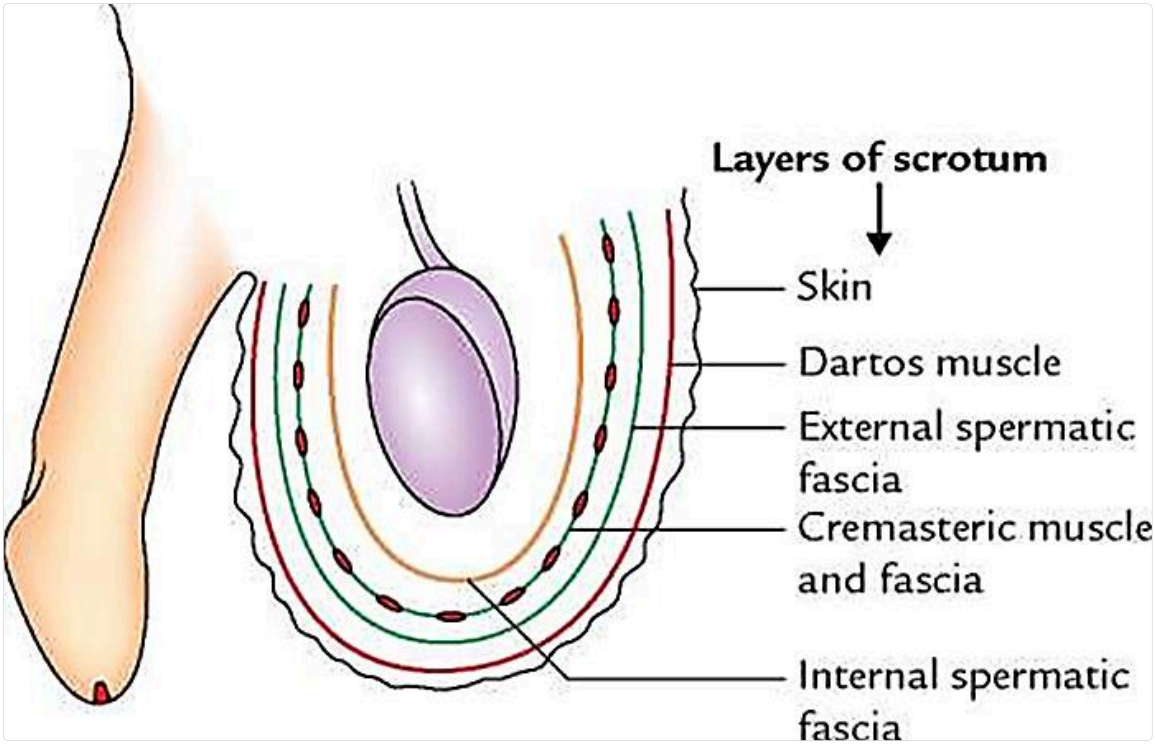


Fig 10.1 — original page 25

Types:

I- Primary hydrocele

II- Secondary hydrocele

I- Primary hydrocele:

When there is no obvious cause to explain collection of fluid in tunica vaginalis sac.

There may be collection of fluid in the tunica vaginalis due to:

- Excessive secretion of fluid within the tunica vaginalis sac
- Defective absorption of the secreted fluid—most primary hydrocele is due to this cause.
- Due to defective lymphatic drainage of the scrotal structures
- Peritoneal fluid may reach the patent processus vaginalis leading to a congenital hydrocele.

II- Secondary hydrocele:

- When there is collection of fluid in tunica vaginalis sac secondary to some disease of testis or the epididymis. Secondary hydrocele are usually small and lax.

- Secondary hydrocele may be secondary to **epididymo-orchitis, testicular**
- **tumor, torsion or trauma.**
- **Q: What is the explanation for development of hydrocele following**

inguinal herniorrhaphy?

One of the explanation is probably due to interruption of lymphatics draining the scrotal contents following herniorrhaphy.

Q: What is hydrocele of the hernial sac?

This is one of the complications of an inguinal hernia. If the neck of the hernial sac is plugged with omentum then there may be collection of fluid in the distal part of the sac resulting in hydrocele of the hernial sac.

Q: What is chylocele?

- A type of hydrocele resulting from lymphatic obstruction of the scrotal contents leads to development of chylocele. One of the lymphatic varix may rupture and discharge chyle into the tunica vaginalis
- This is usually secondary to filarial epididymo-orchitis.
- The fluid contains liquid fat, which is rich in cholesterol.

Q: What is hematocele?

Hemorrhage into the tunica vaginalis may occur due to:

- Tapping of hydrocele
- Trauma to the testis.
- Tumor of the testis

Types of hydroceles:

There are different types of hydrocele depending on the extent of fluid collection in The processus vaginalis:

i. Congenital hydrocele:

- When the whole of processus vaginalis is patent and there is accumulation of fluid within the processus vaginalis.

- The hydrocele sac communicates with the peritoneal cavity through the deep ring. So this is also known as communicating hydrocele.
- The swelling is inguinoscrotal, soft, cystic, fluctuant and also trans-illuminant.

ii. Funicular hydrocele:

- When there is collection of fluid in the patent processus vaginalis extending from the deep inguinal ring to the upper pole of the testis then it is called a funicular type of hydrocele.

iii. Infantile hydrocele:

- There is collection of fluid in the processus vaginalis, which extend from the tunica vaginalis of the testis up into the inguinal canal and extends up to but not beyond the deep inguinal ring.

iv. Encysted hydrocele of the cord:

- When there is accumulation of fluid in the intermediate segment of processus vaginalis, then it is called encysted hydrocele of the cord.

v. Vaginal hydrocele:

- When there is accumulation of fluid in the tunica vaginalis sac then it is called vaginal hydrocele.
- The swelling is scrotal, soft cystic, fluctuant and trans-illuminant.

vi. Bilocular hydrocele or hydrocele en bisac:

- There is accumulation of fluid in the tunica vaginalis sac in the scrotum and there is prolongation of the tunica vaginalis sac into the inguinal.
- There are two loculi, one lying in the scrotum and another in the inguinal canal, which shows cross-fluctuation.

vii. Hydrocele of the hernial sac:

There is accumulation of fluid in the distal part of the hernial sac.

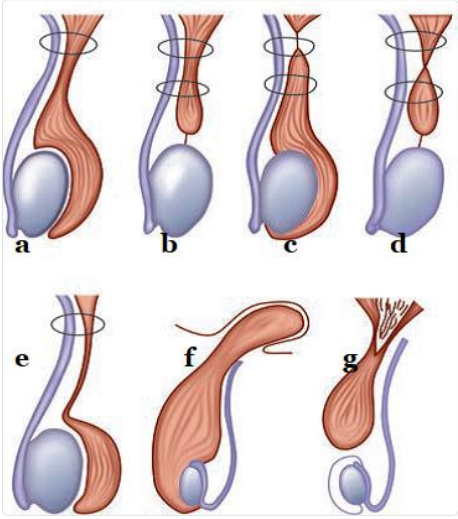


Fig 10.2 — original page 28

Types of hydroceles

Q: What are the characteristics of fluid in hydrocele?

The fluid is usually amber coloured. It contains inorganic salts, traces of albumin and fibrinogen

Clinical picture of vaginal hydrocele:

Symptoms:

- Unilateral or bilateral scrotal swelling
- Usually painless or sometimes sense of heaviness

Signs:

- Unilateral or bilateral symmetrical scrotal swelling
- Normal scrotal skin
- The swelling does not show expansile impulse on cough
- Scrotal neck test: the swelling is pure scrotal swelling
- The swelling is soft cystic in consistency
- Surface is smooth, margins are rounded, non-tender
- **Fluctuation is positive**
- The swelling is held at the upper pole in between the thumb and the index finger of one hand to make the swelling tense.
- These are watching fingers. The thumb and index finger of the other hand is placed apart at the lower pole of the swelling. These are displacing fingers.

- The swelling is now pressed at the lower pole between the thumb and the index finger.
- The fingers at the upper pole will be lifted up due to the pressure exerted at the lower pole.
- This test is repeated at right angle to the previous axis. If the watching fingers are lifted up at both axes then the fluctuation is said to be positive

  .
- Trans-illumination is positive
- No palpable expansile impulse on cough
- Dull on percussion.
- The testis cannot be felt separately from the swelling.

D.D. of hydrocele:

1. Hematocele.
2. Chylocele.
3. Testicular tumor.
4. Epididymal cyst.
5. Inguinal hernia.
6. Encysted hydrocele of the cord. 7. Filariasis of scrotum.

Complications of hydrocele :

- Infection—pyocele
- Trauma—hematocele
- Atrophy of testis
- Rupture
- Calcification of the sac

Investigations:

Scrotal ultra-sound:

- confirm the diagnosis
- exclude other possibilities
- detect state of the testis (? Atrophy)

Treatment:

1- Eversion of tunica vaginalis:

- An incision is made over the tunica vaginalis in an avascular area anteriorly away from the testis, epididymis and cord structures and fluid drained.
- The tunica vaginalis incision is then extended and testis delivered out of the tunica vaginalis sac.
- **Eversion of sac**: The cut margin of the tunica vaginalis sac is everted around the testis.
- **Suturing the cut margins of tunica vaginalis:** The cut margin is stitched behined the testis with absorbable sutures.
- Hemostasis is secured and the testis with the everted sac placed back into the scrotal sac.
- **Closure:** The dartos muscle and skin.

2- Lord's operation:

plication of tunica vaginalis

- In this procedure after incision in the tunica vaginalis sac, the fluid is drained and the testis delivered from the tunica vaginalis sac.
- The cut margin of the tunica vaginalis is then plicated with absorbable sutures.
- The testis with the plicated tunica vaginalis sac is then placed in the scrotum.

Scrotal incision is then closed with drain inside.

Encysted hydrocele of the cord

This is a condition where there is collection of fluid in un-obliterated intermediate portion of the processus vaginalis. The swelling may be situated in the inguinal, inguino- scrotal or scrotal region.

Clinical picture:

Symptoms:

- Painless scrotal swelling

Signs:

- There is a swelling in relation to spermatic cord.
- The swelling is elongated, soft-cystic in feel, fluctuation and trans-illumination tests are positive.
- There is no expansile impulse on cough,
- the swelling is mobile from side to side and from above downward, on traction on the testis the mobility of the swelling becomes restricted.
- **Traction test:** when the testis is pulled downwards gently the swelling

  Moves downwards and becomes less mobile as the swelling is fixed to the spermatic cord

D.D.

1. Cyst of the epididymis.
2. Spermatocele.
3. Lipoma of the cord.
4. Varicocele.

Treatment:

excision of the cyst from the spermatic cord

CHAPTER 11

Epididymal cyst and spermatocele

IN THIS CHAPTER

01 Epididymal cyst

02 On examination

03 Spermatocele

11

Epididymal cyst

- is a cyst of the epididymis arises due to cystic degeneration of the epididymis**.**
  Symptoms: Very slowly growing scrotal swelling

On examination:

- There is a soft cystic swelling in relation to the head of the epididymis.
- The swelling has a smooth or lobulated surface and feels like a bunch of grapes.
- Testis can be felt separately from the swelling.
- The swelling is brilliantly transilluminant as it contain Crystal clear fluid.

D.D.

- Spermatocele.
- Cyst of the appendix of the testis (Cyst of Morgagni).
- Encysted hydrocele of the cord.
- Chronic Epididymo-orchitis.

Investigation:

Scrotal ultrasound

Treatment:

excision of the cyst

Complication of the excision may cause blockage of the sperm conducting epididymis of that side.

Spermatocele

- Is unilocular retention cyst developing from some portion of the epididymis usually the head of the epididymis
- Lies in the head of the epididymis above and behind the upper pole of the testis.
- Soft, cystic in consistency, fluctuation is positive.
- Trans-illumination poorly positive as it contains turbid fluid.

How will you treat spermatocele ?

Large cyst: aspiration, or complete excision through scrotal route.

CHAPTER 12

Varicocele

IN THIS CHAPTER

01 What are the types of varicocele?

02 Why varicocele is common on left side?

03 Clinical picture of primary varicocele

04 Different grades of varicocele

05 What are the effects of varicocele on testicular function?

06 How surgery is helpful in varicocele?

07 What is Palomo's operation?

08 What is inguinal operation for varicocele?

09 What are the complications of varicocele operation?

12

This is a condition of dilatation and tortuosity of pampiniform plexus of veins of the spermatic cord.

Which veins forms the pampiniform plexus?

The pampiniform plexus are the bunch of veins draining the testis, epididymis, vas deferens and the cremester muscle.

How the pampiniform plexus ends?

i. Anastomosing veins from the testis, epididymis, vas deferens and the cremester muscle gives rise to a plexus of veins at upper pole of the testis which are about 15–25 in number.

ii. As the vein ascends, it reduces to 12–15 in number.

iii. At the level of the superficial inguinal ring the number of vein becomes 4–

5. veins

iv. At the level of deep inguinal ring the veins are reduced to 2 in number.

v. The two vein ascends in the retroperitoneum and join to from a single vein which on right side ends at the inferior vena cava obliquely and on left side ends at the left renal vein at right angle.

vi. Near the termination the testicular vein is provided with a valve.

vii. There is communication between the cremesteric veins and the veins draining the vas which provides alternative path of venous drainage from the testis.

viii. The cremesteric veins drains into the inferior epigastric vein.

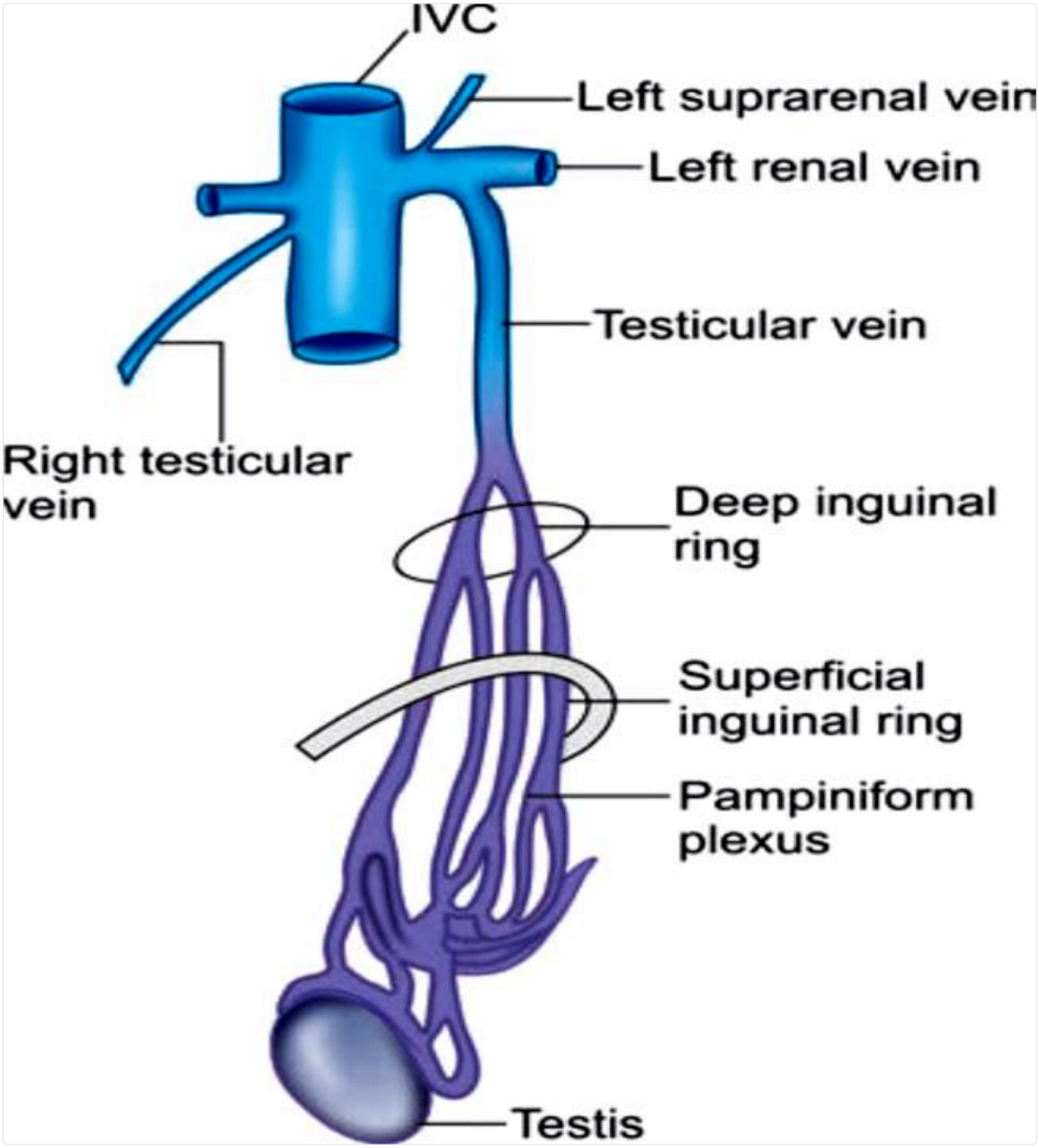


Fig 12.1 — original page 34

What are the types of varicocele?

There are two types of varicocele:

1. Primary:

In 95% cases no cause for varicocele could be found.

2. Secondary:

the varicocele is secondary to obstruction of testicular vein. The obstruction of the testicular vein may be due to retroperitoneal tumor or kidney tumor.

Why varicocele is common on left side?

90% cases of varicocele is found on the left side.

1) The left testicular vein joins the left renal vein at right angle.

2) The left testicular vein joins the left renal vein which drains into the inferior vena cava 8–10 cm more cranial than the right testicular vein, as a result the left testicular vein has a 8–10 cm greater pressure than the right testicular vein.

3) Left testicular vein is longer as left testis lies at a lower level and left renal vein lie at a higher level. Hence, longer column of blood in left testicular vein exerting more pressure in left pampiniform plexus.

4) Left testicular vein is liable to be compressed by loaded pelvic colon as it ascends behind the left colon.

5) The left renal vein may be sandwiched between the abdominal aorta and superior mesenteric artery and compression of the left renal vein may lead to varicocele.

6) Left testicular artery may arch over the left renal vein and may cause compression of the renal vein and testicular vein leading to varicocele.

7) In carcinoma of left kidney the tumor may grow along the renal vein causing obstruction of left testicular vein leading to varicocele.

- **The mechanism of Varicocele formation usually is caused by:**

1. Increased venous pressure in the left renal vein.

2. Incompetent valves of the testicular vein.

Clinical picture of primary varicocele:

a) Swelling in one side commonly the left side of the scrotum.

b) The swelling started in the lower part of the scrotum and subsequently slowly increasing in size and grown up to the root of the scrotum.

c) The swelling disappears on lying down position and reappears on standing and walking.

d) Patient complains of dull aching pain in the affected side of the scrotum which is more during standing and towards the evening when the swelling enlarges in size.

e) Patient may present with infertility

f)On examination:

1. A mass of dilated vein feeling like a bag of worms is palpable on the left side of the scrotum along the left spermatic cord extending from the upper pole of the testis up to the superficial inguinal ring.

2. No expansile impulse on cough is palpable, instead a thrill is palpable.

3. On lying down and on elevation of the scrotum the swelling disappears.

On asking the patient to stand up the dilated vein reappeared.

4. The testicular volume on affected side is smaller than the other one.

5.Bow sign:

The varicocele mass is held between index and middle finger and the thumb and the patient is asked to bow down—the tension within the pampiniform plexus of veins becomes less and the varicocele gets reduced in size. Bowing position cuts off the continuity of the blood inside the varicocele and the testicular vein.

6. Abdominal examination is normal (as it is primary type)

Different grades of varicocele:

Depending on the physical finding there are three grades of varicocele.

Grade I:

Small varicocele which is palpable only when patient performs Valsalva Maneuver (expiration against a closed glottis).

Grade II:

Moderate sized. Easily palpable varicocele without Valsalva's maneuver.

Grade III:

Large varicocele visible through the scrotal skin.

D.D.

1. Inguinal hernia
2. Congenital hydrocele
3. Encysted hydrocele of the cord
4. Lipoma of the cord
5. Epididymal cyst
6. Vaginal hydrocele.

What are the effects of varicocele on testicular function?

a)Hyperthermia:

Presence of varicosities impedes the countercurrent heat exchange mechanism. Increased scrotal temperature inhibits spermatogenesis.

b)Intratesticular hyperperfusion injury:

Abnormal elevation of microvascular blood flow and increased intra-testicular temperature increases the metabolic activity and depletes the intracellular glycogen store and induces testicular parenchymal injury.

c) Hypoxia.

d) Local testicular hormonal imbalance: Leydig cell dysfunction in patients with varicocele may be caused by diminished intra-testicular testosterone.

e) Sperms Maturation arrest f) decreased testicular volume. So assessment of testicular volume is essential in evaluation of varicocele patient.

- **How scrotal temperature is kept lower?**

i. For normal spermatogenesis the scrotal temperature is kept 2.5°C lower than the core temperature.

ii. The pampiniform plexus encircles the testicular artery as it supplies the testis.

iii. There is effective countercurrent heat exchange mechanism that permits the arterial blood to be cooled from intra-abdominal temperature to the cooler temperature found in the scrotum.

Investigations:

1- Color Duplex scan study is helpful for diagnosis of varicocele.

The diagnosis is made when two to three veins of 3 mm or greater in size are found Or there is enlargement of veins and reflux on Valsalva maneuver

2- Semen analysis:

in case of infertility.

3- C.T. scan abdomen for secondary type to detect etiology as hypernephroma

Treatment:

According to the grade:

Grade I varicocele does not require active management.

Grade II and grade III varicocele need treatment when associated with:

1. Subfertility and infertility.

2. Testicular volume loss greater than 2 mL in comparison to the normal side.
3. Pain and discomfort.
4. Cosmetic problems.

How surgery is helpful in varicocele?

1. The stagnation of blood in the pampiniform plexus is prevented.
2. The spermatogenesis may return to normal.
3. The testicular volume may return to normal

Types of surgery for varicocele:

A. High approach = Palomo' operation

B. Low = Inguinal approach

What is Palomo's operation?

- This is a technique of high ligation of testicular vein.
- An oblique incision is made 3 cm above the level of deep inguinal ring.
- After incising the skin, subcutaneous tissue, external oblique muscle and aponeurosis, internal oblique and transversus abdominis muscle the retroperitoneum is exposed.
- Testicular vein is dissected in retroperitoneum lateral to external iliac artery.
- The testicular vein is then ligated and divided.
- The procedure can be done laparoscopically.
- The venous drainage of testis will be **maintained** by anastomosis of pampiniform plexus with the cremesteric veins and the veins draining the vas.

What is inguinal operation for varicocele?

1) Through a groin incision—the spermatic cord is exposed and cremasteric muscle and fascia is incised.

2) The vas deferens and the vessels supplying the vas are separated from the main mass of dilated veins. Two pairs of clamps are applied proximally and distally at a gap of 3 cm and the intervening segment of veins are excised.

3) The two clamps are then approximated and ligated. This removes the main mass of dilated vein and also shortens the cord.

4) At the completion of operation only the testicular artery, lymphatics and the vas deferens with its vessels remain intact.

What are the complications of varicocele operation?

- **Hydrocele:** This is due to lymphatic obstruction.
- **Testicular artery injury:** There is 10–20% chance of testicular atrophy. In children the incidence of testicular atrophy is less than in adults.
- Recurrence of varicocele.

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